THE TOTAL SITE OF A SIMPLICIAL OBJECT IN SITES WITH COCONTINUOUS STRUCTURE FUNCTORS OF PROPERTY P

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author writes 'Compatibility of this type of fibre product with compositions of functors implies the claim (details omitted)' (source lines 875-878) - this is the auxiliary identification W x_{(phi,g),U,(id,f_i)} U_i = W x_{g,U,f_i} U_i, whose mechanism is the compatibility property of the Sites remark cited in the statement; accepted per the auxiliary-omission precedent of cards L46160, L46162 and L46170; the covering stability itself is delivered by property P.

Editorial note. The label lemma-simplicial-cocontinuous-site names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The simplicial object in sites with cocontinuous functors. Same chapter, Situation \label{situation-simplicial-site} (cocontinuous variant) and Sites, Remark \label{sites-remark-cartesian-cocontinuous}. Statement lines 844-848: the functors u_phi : C_n -> C_m are cocontinuous with property P, which guarantees the fibre products W x_{g,U,f_i} U_i in C_m and the covering stability.

Source reference. Original remark 09W7: sources/stacks-chow/tags.txt, source lines 12485–12485. Prerequisite pointer invoked at proof line(s) 848; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a simplicial object in the category whose objects are sites and whose morphisms are cocontinuous functors. With notation as above, assume the functors $u_\varphi : \mathcal{C}_n \to \mathcal{C}_m$ have property P of Sites, Remark 09W7. Then we can construct a site $\mathcal{C}_{total}$ as follows.*

1. *An object of $\mathcal{C}_{total}$ is an object U of $\mathcal{C}_n$ for some n,*
2. *a morphism $(\varphi, f) : U \to V$ of $\mathcal{C}_{total}$ is given by a map $\varphi : [m] \to [n]$ with $U \in \mathrm{Ob}(\mathcal{C}_n)$, $V \in \mathrm{Ob}(\mathcal{C}_m)$ and a morphism $f : u_\varphi(U) \to V$ of $\mathcal{C}_m$, and*
3. *a covering $\{(id, f_i) : U_i \to U\}$ in $\mathcal{C}_{total}$ is given by an n and a covering $\{f_i : U_i \to U\}$ of $\mathcal{C}_n$.*

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Proof. Composition of $(\varphi, f) : U \to V$ with $(\psi, g) : V \to W$ is given by $(\varphi \circ \psi, g \circ u_\psi(f))$. This uses that $u_\psi \circ u_\varphi = u_{\varphi\circ\psi}$.

Let $\{(\mathrm{id}, f_i) : U_i \to U\}$ be a covering as in (3) and let $(\varphi, g) : W \to U$ be a morphism with $W \in \mathrm{Ob}(\mathcal{C}_m)$. We claim that

$$W \times_{(\varphi,g),U,(\mathrm{id},f_i)} U_i = W \times_{g,U,f_i} U_i$$

in the category $\mathcal{C}_{total}$ where the right hand side is the object of $\mathcal{C}_m$ defined in Sites, Remark 09W7 which exists by property P. Compatibility of this type of fibre product with compositions of functors implies the claim (details omitted). Since the family $\{W \times_{g,U,f_i} U_i \to W\}$ is a covering of $\mathcal{C}_m$ by property P we see that the collection of coverings is stable under base change. The other axioms of a site are immediate. $\square$